Aditya Saini

908-912-6974 | adityasaini0328@gmail.com | linkedin.com/in/adityavsaini | github.com/Saini-01 | sunnyv.surge.sh

Education

Purdue University West Lafayette, IN

B.S. Computer Science (Systems Programming) *Expected Graduation: December 2028*

Relevant Coursework: Computer Architecture, Programming in C, Data Structures & Algorithms, Object-Oriented Programming, Discrete Mathematics, Linear Algebra, Multivariate Calculus

Experience

Software Engineer Intern Jan. 2027 – Aug. 2027

RTX: Collins Aerospace *Cedar Rapids, IA*

- Incoming Software Engineer Intern at Collins Aerospace on the Avionics Software Tools team.

Embedded Engineer Sep. 2025 – Present

Purdue Electric Racing *West Lafayette, IN*

- Designed and implemented the driver-facing dashboard UI for the Electric Racing competition vehicle on a Nextion HMI display using Nextion Editor, authoring all screen graphics for live telemetry, fault codes, and sensor data
- Wrote embedded C on an STM32F4 (ARM Cortex-M4) driving 6+ dashboard screens and button-mapped page transitions over UART, mapping CAN fault codes to prioritized driver alerts
- Rewrote the display command queue from fixed-width slots to a packed byte-level ring buffer, eliminating padding on the UART stream and fixing intermittent dropped commands on the Nextion dashboard display

Robotic Safety Research Assistant January 2026 – Present

PurSec, Purdue Computer Science *West Lafayette, IN*

- Researched safety of multimodal LLMs controlling embodied agents, formulating attack methods to create physical harm from benign individual steps for over 20 individual attacks spanning 8 different physical attack categories
- Built a closed-loop harness through a state machine in Python that returns updated scene state and full action history to the verifier after every robot step, gating each proposed action on both safety brains before execution
- Built the Python simulation/eval pipeline using AI2-THOR rendering, Pillow + FFmpeg video output, and a Dockerized Ubuntu 22.04 + Xvfb headless harness for reproducible runs on AWS EC2 servers.

Projects

Natural Language Shell | *C, POSIX, Python, Bash* June – August 2025

- Built NLSH, a custom Unix shell in C with process management, input tokenization, and shell built-ins (cd, help, exit), handling the full read-parse-execute loop rather than wrapping an existing shell
- Bridged the C core to a Python helper over popen for command translation, added OS-aware prompt context, and shipped a Bash installer that puts the shell on PATH

FRC 2554: Build President & Programming Captain | *Python, OpenCV, CUDA* Sep. 2021 – Aug. 2025

- Led 60+ students as Build President to District Championships after 18 years, earning 3 regional awards.
- Built a CUDA-accelerated OpenCV vision pipeline in Python for AprilTag detection, improving autonomous positioning accuracy by 66%, leading to a larger scoring volume with greater scoring efficiency.
- Diagnosed and resolved cross-system hardware failures across CAN bus-chained motor controllers and sensors
- Designed robot subsystems in Onshape through successive prototyping cycles, including the intake and drivetrain

Custom 30-Button Gamepad | *Embedded C, PSoC, USB* January – April 2025

- Fabricated a custom 30-button gamepad on a PSoC microcontroller, writing embedded C firmware and a USB polling loop so the FRC Driver Station read it as a standard game controller and relayed inputs to the robot
- Improved driver control efficiency by 50% by giving every robot function its own dedicated button

Technical Skills

Languages: C, C++, SystemVerilog, Verilog, RISC-V assembly, Python, Java, SQL, Bash
Systems: Linux, POSIX, process management, embedded C, RTOS, Docker, Makefiles, GCC, GDB
Embedded/Hardware: STM32 (Cortex-M4), STM32CubeIDE, PSoC, Jetson Orin Nano, UART, CAN, SPI, I2C
EDA/Tools: Cadence Genus/Innovus, AWS, OpenLane, LTspice, FPGA synthesis/timing, RTL verification, CUDA, Git